c) Backward Chaining

Aim:

To implement Backward Chaining (goal-driven reasoning) using PROLOG.

Procedure:

1. Open Notepad (or the SWI-Prolog editor).
2. Write the facts and rules and save the file with **.pl** extension.
3. In SWI-Prolog, load the program using **File -> Consult** and select the file.
4. Type the query at the **?-** prompt, ending it with a full stop (.). Use **;** to get further answers.
5. Observe and record the output.

Prolog Code:

Exercise 1: Medical Diagnosis

```
% Facts
has_fever(john).
has_cough(john).

% Rule
has_flu(X) :- has_fever(X), has_cough(X).

% Query
% ?- has_flu(john).
```

Output:

true.

Exercise 2: Student Eligibility Checker

```
% Facts
passed(alice, maths).
passed(alice, science).

% Rule
eligible(X) :- passed(X, maths), passed(X, science).

% Query
% ?- eligible(alice).
```

Output:

true.

Exercise 3: Travel Planner

```
% Facts
has_ticket(james).
has_visa(james).

% Rule
can_travel(X) :- has_ticket(X), has_visa(X).

% Query
% ?- can_travel(james).
```

Output:

```
true.
```

Result:

Thus Backward Chaining was implemented using PROLOG and executed successfully.